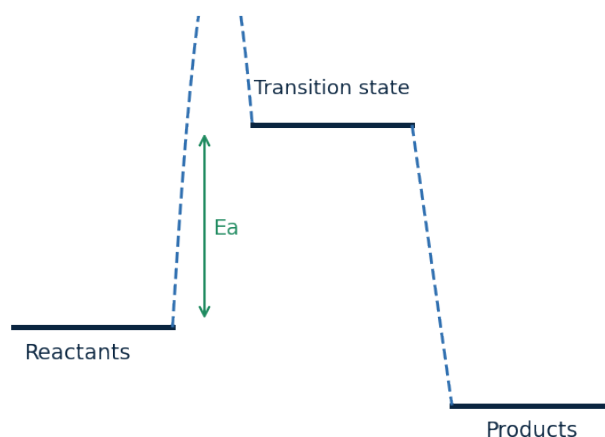


Thermodynamics

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Thermodynamics deals with energy changes accompanying chemical and physical processes. A system can be open, closed, or isolated depending on exchange of matter/energy with surroundings.
- First Law: $\Delta U = q + w$ (internal energy change equals heat added plus work done on the system). Internal energy is a state function.
- Enthalpy $H = U + PV$. At constant pressure, $\Delta H = q_p$, the heat absorbed/released by the reaction.
- Exothermic reactions release heat (ΔH negative); endothermic reactions absorb heat (ΔH positive).
- Hess's Law: the total enthalpy change for a reaction is the same regardless of the number of steps taken (enthalpy is a state function), allowing indirect calculation via known reactions.
- Entropy (S) is a measure of disorder/randomness. Second law: for a spontaneous process, the total entropy of system + surroundings increases.
- Gibbs free energy: $\Delta G = \Delta H - T \Delta S$. A process is spontaneous if $\Delta G < 0$ at constant temperature and pressure.
- Standard enthalpy of formation (ΔH_f) is the enthalpy change when 1 mole of a compound forms from its elements in their standard states.



Reaction energy profile showing the activation energy (E_a) between reactants and products.

2. Formula Sheet

First law of thermodynamics	$\Delta U = q + w$
Enthalpy	$H = U + PV$
Enthalpy at constant P	$\Delta H = q_p$
Gibbs free energy	$\Delta G = \Delta H - T \Delta S$
Hess's law (enthalpy of reaction)	$\Delta H(\text{rxn}) = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$
Work done (gas expansion)	$w = -P \Delta V$
Spontaneity condition	$\Delta G < 0 \Rightarrow \text{spontaneous}$
Relation between ΔH and ΔU	$\Delta H = \Delta U + \Delta n(\text{gas}) RT$

3. Practice Problems (22 questions)

1. A reaction releases 100 kJ of heat and does 20 kJ of work on the surroundings. Find the change in internal energy of the system.
2. Find $\Delta H(\text{reaction})$ for $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$, given $\Delta H_f(\text{CO}_2) = -393.5 \text{ kJ/mol}$ (ΔH_f of elements = 0).
3. State whether the following reaction is exothermic or endothermic: $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$, $\Delta H = +180 \text{ kJ}$. Explain your reasoning.
4. A process has $\Delta H = -50 \text{ kJ}$ and $\Delta S = -100 \text{ J/K}$ at 300 K. Find ΔG and state whether the process is spontaneous.
5. State Hess's law and explain why it allows enthalpy of a multistep reaction to be found by adding known reactions.
6. Find ΔH for a reaction that is the reverse of one with $\Delta H = -250 \text{ kJ/mol}$.
7. Explain why entropy generally increases when a solid melts into a liquid.
8. For the reaction $\text{A(g)} \rightarrow 2\text{B(g)}$, $\Delta U = 30 \text{ kJ}$ at 300 K. Find ΔH ($R = 8.314 \text{ J/mol K}$).
9. Explain what is meant by a 'state function' and give two examples.
10. A gas expands against a constant external pressure of 2 atm, and its volume increases by 3 L. Find the work done by the gas ($1 \text{ L atm} = 101.3 \text{ J}$).
11. State the second law of thermodynamics in terms of entropy change of the universe.
12. Explain why $\Delta G < 0$ is the criterion for spontaneity rather than $\Delta H < 0$ alone.
13. Find the standard enthalpy of reaction for $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, given $\Delta H_f(\text{CH}_4) = -74.8$, $\Delta H_f(\text{CO}_2) = -393.5$, $\Delta H_f(\text{H}_2\text{O}) = -285.8 \text{ kJ/mol}$.
14. Explain the difference between an exothermic and an endothermic process in terms of the sign of ΔH .
15. A reaction has $\Delta H = +40 \text{ kJ}$ and $\Delta S = +100 \text{ J/K}$. Find the temperature above which the reaction becomes spontaneous.

16. State the first law of thermodynamics and explain the sign convention used for heat and work.
17. Explain why entropy of the universe increases for a spontaneous irreversible process.
18. Find ΔU for a process at constant volume where $q = 500 \text{ J}$ is absorbed by the system (no work done).
19. Explain why melting of ice at 0°C is spontaneous even though ΔH is positive (endothermic).
20. State the difference between a state function and a path function, giving one example of each.
21. For a reaction, $\Delta H = -120 \text{ kJ/mol}$ and $\Delta S = -300 \text{ J/(mol K)}$. Find ΔG at 400 K , and state if it's spontaneous at that temperature.
22. Explain the meaning of standard enthalpy of formation, and state its value for an element in its standard state.

Answer Key

1. $\Delta U = 80 \text{ kJ}$ (heat released by system is -100 kJ from system's perspective, work done on surroundings means $w = -20 \text{ kJ}$ for system; $\Delta U = -100 + (-20) = -120 \text{ kJ}$, revise: using $q = -100 \text{ kJ}$, $w = -20 \text{ kJ}$ (work done BY system so w is negative for the system) $\Rightarrow \Delta U = -120 \text{ kJ}$)
2. $\Delta H = -393.5 \text{ kJ/mol}$
3. Endothermic (positive ΔH means heat is absorbed)
4. $\Delta G = -50000 - 300(-100) = -20000 \text{ J} = -20 \text{ kJ}$, spontaneous ($\Delta G < 0$)
5. Conceptual - enthalpy is a state function, so total change is path-independent; known step reactions can be summed
6. $\Delta H = +250 \text{ kJ/mol}$
7. Conceptual - liquid state has more disorder/freedom of movement than the ordered solid lattice, increasing entropy
8. $\Delta H = 30000 + (1)(8.314)(300) = 32494 \text{ J}$ approx 32.5 kJ
9. Conceptual - depends only on initial/final state, not path; e.g., internal energy (U), enthalpy (H)
10. $w = -607.8 \text{ J}$ (work done by gas, so negative from system's perspective)
11. Conceptual - total entropy of system + surroundings increases for any spontaneous process
12. Conceptual - ΔG accounts for both enthalpy and entropy changes (via $T \Delta S$), giving the true spontaneity criterion
13. $\Delta H(\text{rxn}) = -393.5 + 2(-285.8) - (-74.8) = -890.3 \text{ kJ/mol}$
14. Conceptual - exothermic: ΔH negative (heat released); endothermic: ΔH positive (heat absorbed)
15. $T = \Delta H / \Delta S = 40000/100 = 400 \text{ K}$
16. Conceptual - ΔU positive for heat added to system; negative work convention varies ($w =$ work done ON system, positive if compressed)
17. Conceptual - overall entropy (system + surroundings) increases even if the system's own entropy decreases locally
18. $\Delta U = +500 \text{ J}$
19. Conceptual - overall entropy increases significantly (ice to water), outweighing the unfavorable positive ΔH at 0°C , giving negative ΔG
20. Conceptual - state function depends only on state (e.g., enthalpy); path function depends on path taken (e.g., heat, work)

21. $\Delta G = -120000 - 400 \times (-300) = 0 \text{ kJ}$ (borderline, at equilibrium at exactly 400K)

22. Conceptual - enthalpy change when 1 mole of compound forms from elements in standard states; zero for elements in their standard state

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.